Per-Layer Misdetection Probability in Hierarchical Beam Search: Channel Model, Exact Formula, and High-SNR Approximation

Draft note

Abstract

This note derives a closed-form expression for the per-layer error probability $P_e(N, \theta, \mathrm{SNR})$ in hierarchical beam search under Rician fading. Starting from the received signal model (Part 1), we obtain an exact binomial sum via the Laplace transform of the correct-beam power distribution (Part 3), and then derive the harmonic-number approximation that exposes the $\ln N$ scaling reported in the companion paper (Part 3).

Contents

1 Part 1: Channel model and signal statistics **2**
1.1 Core assumptions 2
1.2 Marginal distribution of the measured power 4

2 Part 2: Error probability via the Laplace transform **5**

3 Part 3: High-SNR approximation and the $\ln N$ scaling **6**
3.1 Per-layer error probability approximation 6
3.2 Reliability constraint 8
3.3 Error probability approximation quality 9

1 Part 1: Channel model and signal statistics

1.1 Core assumptions

At layer ℓ of the hierarchical search, the device sweeps N candidate sectorized beams that together tile a sector of angular width θ (radians). The true angle-of-arrival (AoA) φ falls inside exactly one of these beams, which is the *correct beam* k^*. The device measures the received power on every beam sequentially, then selects the beam with the highest measured power. A *layer error* occurs when the selected beam differs from k^*.

Received signal model: For beam k the received complex baseband signal is

$$y_k \;=\; \sqrt{G_k\, P_{\mathrm{rx}}}\; h_k \;+\; n_k, \tag{1}$$

where P_{rx} is the mean received power in the absence of beamforming gain (determined by the link budget and path loss), G_k is the beamforming gain for beam k, h_k is the complex fading coefficient, and n_k is receiver noise.
Consider a uniform linear array (ULA) with N_{ant} physical elements.

Assumption 1. Boundary-centered beam placement. The N beam centers are placed at the edges and interior of the sector with equal angular spacing:

$$\phi_k = \phi_{\min} + k \cdot \frac{\theta}{N-1}, \qquad k = 0, 1, \ldots, N-1, \tag{2}$$

so $\phi_0 = \phi_{\min}$ and $\phi_{N-1} = \phi_{\max}$ lie exactly on the sector boundaries. The HPBW of each beam is set equal to the inter-beam spacing $\theta/(N-1)$ so the $-3\,\mathrm{dB}$ points of adjacent beams coincide at the midpoints between centers. Each beam extends $\mathrm{HPBW}/2 = \theta/(2(N-1))$ beyond its center, and the sector is therefore over-covered by this amount on each side: the effective covered arc is $[\phi_{\min} - \theta/(2(N-1)),\ \phi_{\max} + \theta/(2(N-1))]$.

In hierarchical search, the sector at layer ℓ is defined by the selected beam at layer $\ell-1$, whose $-3\,\mathrm{dB}$ points become the new sector boundaries. An AoA near the sector edge therefore had degraded SNR at the previous layer and is the most likely location of a prior misdetection. Placing beam centers exactly on the boundaries ensures that these worst-case AoA positions are covered at peak gain, rather than at the $-3\,\mathrm{dB}$ point as would occur with interior placement (θ/N spacing with no beam center on the boundary).

For a ULA the HPBW is $101.5^\circ/n$ when n elements are active, so equating HPBW to $\theta/(N-1)$ gives the required number of active elements $n(N,\theta) = \frac{101.5^\circ\,(N-1)}{\theta}$, capped at the physical array size N_{ant}. The beamforming gain on the target beam is therefore

$$G(N,\theta) = \min\!\left(N_{\mathrm{ant}},\ \frac{101.5^\circ\,(N-1)}{\theta}\right), \tag{3}$$

and the gain on every non-target beam is zero, $G_j = 0$ for $j \neq k^*$. The maximum number of beams the array can support over sector θ is reached when $n(N,\theta) = N_{\mathrm{ant}}$:

$$N_{\max}(\theta) = \left\lfloor \frac{N_{\mathrm{ant}}\,\theta}{101.5^\circ} \right\rfloor + 1. \tag{4}$$

The gain G depends jointly on N and θ: more beams over the same sector means narrower beams, more active elements, and higher gain. For $N > N_{\max}$, the required HPBW is narrower than the full array can form, which also causes coverage gaps.

Remark. Zero gain on non-target beams is a mathematical idealization (ideal rectangular beam patterns, no sidelobes). Real codebooks have non-zero sidelobes, so $G_j > 0$ for $j \neq k^*$ in practice, raising the floor of the error probability. The assumption yields a lower bound on P_e; any sidelobe energy can make misdetection more likely.

Assumption 2. Rician single-path channel. The fading coefficient on the correct beam is

$$h_{k^*} = \sqrt{\frac{K}{K+1}} + \sqrt{\frac{1}{K+1}}\, w, \qquad w \sim \mathcal{CN}(0,1), \tag{5}$$

where $K \geq 0$ is the Rician K-factor (ratio of LOS power to scatter power). The scatter component w is independent across beams and across time slots.

The Rician model captures the two dominant propagation mechanisms at mmWave frequencies: a strong specular LOS component (weighted by $K/(K+1)$) and an aggregate of weak diffuse reflections modelled as circularly symmetric Gaussian noise. For the simulations, $K = 10$ (i.e. the LOS component carries 90.9% of the power), which is representative of indoor or short-range outdoor links at 28 GHz.

Remark: A single-path model ignores clustered multipath. In practice, several dominant reflectors may each produce a beam-scale peak, and the strongest measured beam may not correspond to the desired LOS direction. Independence across time slots (block fading) is standard but slightly optimistic. Temporally correlated fading could correlate the noise on successive beam measurements, which our analysis does not consider.

Assumption 3. AWGN receiver noise. The noise sample on each beam measurement is $n_k \sim \mathcal{CN}(0, N_0)$, i.i.d. across beams and time slots, where $N_0 = kT_0B \cdot 10^{N_F/10}$ is the effective noise power spectral density including receiver noise figure N_F.

At the symbol level, thermal noise is well modeled as white Gaussian. Independence across beam measurements holds as long as the probe duration is much shorter than the channel coherence time (so the channel does not change). The noise realizations are independent (guaranteed for non-overlapping pilot symbols in OFDM).

Remark: Phase noise from the local oscillator introduces correlated perturbations across measurements. degrades. Hardware non-linearity also creates measurement-dependent biases. Neither effect is considered here, incorporating them would require a non-white noise model.

Assumption 4. Uniform AoA prior. The true AoA φ is uniformly distributed over $[-\pi, \pi)$.

Remark. In practice, the device may have coarse prior information (e.g., from GPS, previous beam

management, or a coarse layer-0 scan, e.g., via SSB in 5G NR). This would change the prior density over the swept sector and the scale but not the core tradeoffs.

Assumption 5. Layer independence. Each hierarchical layer is treated as a fresh beam selection problem, i.e., the sector to sweep at layer ℓ is determined by the outcome of layer $\ell - 1$, but fading, noise, and AoA within the sector are independent of all previous layers.

Remark. This assumption holds when consecutive layer probes are separated by more than the channel coherence time, or when different pilot resources (frequency, code) are used per layer so that noise samples are independent. If all layers are sounded within one coherence interval, fading is correlated across layers: a noise spike that caused an error at layer ℓ will tend to recur at layer $\ell + 1$, which makes errors *positively* correlated and means the true multi-layer success probability $\prod_\ell (1 - P_e^{(\ell)})$ is an approximation. In this case, the per-layer formula derived below would remain valid, but the independence across the product would be lost. In practice, however, layers ℓ and $\ell + 1$ are separated by waiting time.

1.2 Marginal distribution of the measured power

Let $Y_k = |y_k|^2$ be the measured power on beam k.

Target beam. Under Assumptions 1–3 the received amplitude on the target (correct) beam is

$$y_{k^*} = \sqrt{G\,P_{\mathrm{rx}}}\,h_{k^*} + n_{k^*}.$$

Assuming $h_{k^*} \sim \mathcal{CN}(\mu, \sigma_h^2)$ (Rician fading) and n_{k^*} is independent complex Gaussian, their sum remains complex Gaussian, and the envelope $|y_{k^*}|$ follows a Rician distribution. Consequently, $Y_{k^*} = |y_{k^*}|^2$ follows a *scaled noncentral chi-squared* distribution with two degrees of freedom with PDF

$$f_{Y_{k^*}}(y) = \frac{1}{\sigma_{\mathrm{tot}}^2}\,\exp\!\left(-\frac{\nu + y}{\sigma_{\mathrm{tot}}^2}\right) I_0\!\left(\frac{2\sqrt{\nu\,y}}{\sigma_{\mathrm{tot}}^2}\right), \quad y \geq 0, \tag{6}$$

where $\nu = \frac{G\,P_{\mathrm{rx}}\,K}{K+1}$ is the LOS power and $\sigma_{\mathrm{tot}}^2 = \frac{G\,P_{\mathrm{rx}}}{K+1} + N_0$ is the total variance, including scatter fading and receiver noise, $G = G(N, \theta)$ as defined in (3), and I_0 denotes the modified Bessel function of order zero.

Remark. Both ν and σ_{tot}^2 depend on (N, θ) through the gain G. As N increases over a fixed sector, G grows (more active elements, narrower beam), so the LOS power ν and the signal variance both increase, improving the SNR on the correct beam.

Non-target beams. Under Assumption 1, $G_j = 0$ for $j \neq k^*$, so $y_j = n_j \sim \mathcal{CN}(0, N_0)$. Therefore,

$$Y_j \sim \mathrm{Exp}(N_0), \qquad j \neq k^*, \tag{7}$$

i.e., each non-target-beam measurement contains only noise and is exponentially distributed with mean N_0. The powers of the $N - 1$ non-target beams are i.i.d.

2 Part 2: Error probability via the Laplace transform

Error event. An error at this layer occurs when at least one wrong beam has higher measured power than the correct beam:

$$\mathcal{E} = \left\{ \max_{j \neq k^*} Y_j > Y_{k^*} \right\}. \tag{8}$$

Per-layer error probability. We define the per-layer error probability as

$$P_e(N, \theta, \sigma^2) = \mathbb{P}\left(\max_{j \neq k^*} Y_j > Y_{k^*} \right), \tag{9}$$

which is implicitly conditional on the true AoA lying within the swept sector of width θ. Under Assumption 4, the probability that φ falls within the swept sector of width θ is $\theta/(2\pi)$.
The $N-1$ powers of non-target beams are i.i.d. and follow $\mathrm{Exp}(N_0)$; therefore their maximum has CDF

$$F_{\max}(y) = \mathbb{P}\left(\max_{j \neq k^*} Y_j \leq y \right) = \left(1 - e^{-y/N_0}\right)^{N-1}. \tag{10}$$

Conditioning on $Y_{k^*} = y$ and averaging over its distribution:

$$P_e(N, \theta) = 1 - \mathbb{E}_{Y_{k^*}}\left[\left(1 - e^{-Y_{k^*}/N_0}\right)^{N-1} \right] = 1 - \int_0^\infty \left(1 - e^{-y/N_0}\right)^{N-1} f_{Y_{k^*}}(y)\, \mathrm{d}y. \tag{11}$$

The integrand is a polynomial in e^{-y/N_0}, which makes it natural to expand the binomial factor as

$$\left(1 - e^{-y/N_0}\right)^{N-1} = \sum_{m=0}^{N-1} \binom{N-1}{m} (-1)^m\, e^{-my/N_0}. \tag{12}$$

Substituting into (11), we obtain

$$P_e(N, \theta) = 1 - \sum_{m=0}^{N-1} \binom{N-1}{m} (-1)^m \underbrace{\int_0^\infty e^{-my/N_0}\, f_{Y_{k^*}}(y)\, \mathrm{d}y}_{= \mathcal{L}_{Y_{k^*}}(m/N_0)}, \tag{13}$$

where $\mathcal{L}_{Y_{k^*}}(s) = \mathbb{E}[e^{-sY_{k^*}}]$ is the Laplace transform of the target-beam power.

Laplace transform of the correct-beam power. Based on the PDF (6) with parameters ν and σ^2_{tot}, the Laplace transform is

$$\mathcal{L}_{Y_{k^*}}(s) = \frac{1}{1 + s\sigma^2_{\mathrm{tot}}} \exp\left(-\frac{\nu s}{1 + s\sigma^2_{\mathrm{tot}}} \right). \tag{14}$$

Setting $s = m/N_0$ in (14) and substituting σ^2_{tot} and ν from (6), we obtain

$$\mathcal{L}_{Y_{k^*}}\left(\frac{m}{N_0} \right) = \frac{1}{1 + m\sigma^2_{\mathrm{tot}}/N_0} \exp\left(-\frac{m\nu/N_0}{1 + m\sigma^2_{\mathrm{tot}}/N_0} \right). \tag{15}$$

Assembling the binomial expansion (13) and (15):

Proposition 1. ***Exact per-layer error probability.*** *Under Assumptions 1–5, the per-layer error probability when sweeping N beams over a sector of width θ is*

$$\boxed{P_e(N,\theta) = 1 - \sum_{m=0}^{N-1} \binom{N-1}{m} (-1)^m \frac{1}{1 + m\sigma_{\text{tot}}^2/N_0} \exp\left(-\frac{m\nu/N_0}{1 + m\sigma_{\text{tot}}^2/N_0}\right)} \tag{16}$$

with ν and σ_{tot}^2 as defined in (6), which in turn depend on $G(N,\theta)$ via (3). This expression is exact for all SNR and all $K \geq 0$.

Remark. This expression confirms $P_e \to 0$ as $N_0 \to 0$ (noiseless limit). Setting $K = 0$ gives $\nu = 0$ and $\sigma_{\text{tot}}^2 = NP_{\text{rx}} + N_0$, so the exponential factor in (16) vanishes and the sum reduces to

$$P_e\big|_{K=0} = 1 - \sum_{m=0}^{N-1} \binom{N-1}{m} \frac{(-1)^m}{1 + mc}, \qquad c = \frac{\sigma_{\text{tot}}^2}{N_0} = 1 + \frac{G(N,\theta)P_{\text{rx}}}{N_0} = 1 + G(N,\theta) \cdot \text{SNR}, \tag{17}$$

which evaluates via the identity $\sum_{m=0}^{n} \binom{n}{m}(-1)^m/(1+mc) = c^n\, n! \,/\, \prod_{k=1}^{n}(1+kc)$ to

$$P_e\big|_{K=0} = 1 - \frac{c^{N-1}\,(N-1)!}{\prod_{k=1}^{N-1}(1+kc)}. \tag{18}$$

3 Part 3: High-SNR approximation and the $\ln N$ scaling

3.1 Per-layer error probability approximation

Proposition 1 is numerically exact but involves an alternating sum of N terms that grows expensive for large N and obscures how error probability depends on the search parameters. The goal here is to obtain an approximation that is both computationally cheap and analytically transparent.

Assumption 6. Moderate-to-high SNR on the correct beam. The effective SNR on the target beam after beamforming, $G_{k^*} \cdot \text{SNR} = G(N,\theta) \cdot P_{\text{rx}}/N_0$, satisfies

$$G(N,\theta) \cdot \text{SNR} \;\gg\; K + 1, \tag{19}$$

where $\text{SNR} = P_{\text{rx}}/N_0$ is the per-element SNR (before beamforming gain) and $G(N,\theta)$ is given by (3).

This is the regime in which beam search is intended to operate. If the per-target-beam SNR were below the noise, beam selection would be essentially random regardless of the algorithm.

Example. For the companion simulations at 28 GHz, $P_{\text{rx}} \approx -90\,\text{dBm}$ (20 m free-space, 10 dBm transmit), noise power $\approx -119\,\text{dBm}$ (100 MHz, 5 dB NF), giving $\text{SNR} \approx 29\,\text{dB}$. The beamformed SNR condition (19) becomes $101.5°(N-1)\cdot\text{SNR}/\theta \gg K+1$, i.e., $(N-1)/\theta \gg (K+1)/(101.5°\cdot\text{SNR})$. For $K = 10$ and $\text{SNR} = 29\,\text{dB}$ (linear ≈ 794), the right-hand side is $11/(101.5° \cdot 794) \approx 1.4 \times 10^{-4}$ deg^{-1}, so the condition is satisfied for any (N,θ) with $(N-1)/\theta \gtrsim 0.01\,\text{deg}^{-1}$, e.g. $N \geq 2$ for $\theta \leq 70°$, or $N \geq 3$ for $\theta \leq 140°$. For wide sectors with few beams (e.g. $N = 2$, $\theta = 120°$, giving $G \approx 0.85 < 1$) the array cannot form a beam narrow enough to achieve meaningful gain, and the exact formula (16) should be used directly.

Remark. At very low SNR (below $K+1$ in linear units) the approximation overestimates the number of relevant terms in the binomial sum, producing rather an optimistic error probability. The exact formula (16) should be used in that regime.

Under Assumption 6, for $m \geq 1$ and $G \cdot \mathrm{SNR} \gg K+1$:

$$\frac{K+1}{K+1+mG\cdot\mathrm{SNR}} \approx \frac{1}{m}\frac{K+1}{G\cdot\mathrm{SNR}}, \tag{20}$$

$$\exp\left(-\frac{KmG\cdot\mathrm{SNR}}{K+1+mG\cdot\mathrm{SNR}}\right) \approx \exp(-K). \tag{21}$$

Substituting into the binomial sum, we obtain:

$$1-P_e(N,\theta) = \sum_{m=0}^{N-1}\binom{N-1}{m}(-1)^m\mathcal{L}_{Y_{k^*}}\left(\frac{m}{N_0}\right) \approx 1+\sum_{m=1}^{N-1}\binom{N-1}{m}(-1)^m\frac{(K+1)e^{-K}}{mG\cdot\mathrm{SNR}}. \tag{22}$$

The key identity needed to evaluate the remaining sum is $\sum_{m=1}^{n}\binom{n}{m}\frac{(-1)^{m+1}}{m}$, which equals to $H_n = \sum_{k=1}^{n} 1/k$, i.e., the n-th harmonic number.
Applying H_n with $n = N-1$ to (22), we obtain

$$\sum_{m=1}^{N-1}\binom{N-1}{m}\frac{(-1)^m}{m} = -H_{N-1}. \tag{23}$$

The per-layer error probability under the approximation becomes

$$P_e(N,\theta) \approx 1-\left[1-\frac{(K+1)e^{-K}}{G(N,\theta)\cdot\mathrm{SNR}}H_{N-1}\right] = \frac{(K+1)e^{-K}H_{N-1}}{G(N,\theta)\cdot\mathrm{SNR}}. \tag{24}$$

Remark. The harmonic number $H_{N-1} = \sum_{k=1}^{N-1} 1/k$ can be further approximated. For $N \geq 2$, the Euler–Maclaurin formula gives $H_{N-1} = \ln(N-1)+\gamma_E+O(1/N)$, where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant. For ranges relevant to beam search, it would be possible to apply the scaling $H_{N-1} \approx \ln N, \quad N \gg 1$, but the approximation is not tight until $N \gtrsim 30$, and H_{N-1} is retained exactly below.

Proposition 2. ***Approximation of per-layer error probability*** *Under Assumptions 1–6, the per-layer error probability is approximated by*

$$\boxed{P_e(N,\theta) \;\approx\; \frac{(K+1)\,e^{-K}\,H_{N-1}}{G(N,\theta)\cdot SNR},} \tag{25}$$

where $H_{N-1} = \sum_{k=1}^{N-1}\frac{1}{k}$ is the $(N-1)$-th harmonic number, $SNR = P_{\mathrm{rx}}/N_0$ is the per-element SNR, and $G(N,\theta) = \min(N_{\mathrm{ant}},\, 101.5^\circ(N-1)/\theta)$ is the beamforming gain on the correct beam from (3).

The error probability decreases with N through two competing effects: H_{N-1} grows (more noise opportunities) while G also grows (stronger signal), with the gain effect dominating for $N \leq N_{\max}(\theta)$.

3.2 Reliability constraint

Proposition 3 yields a tractable per-layer bound that can be inserted directly into the multi-layer reliability constraint of the companion paper. Assuming layer independence (Assumption 5), the overall success probability across L layers satisfies

$$P_{\text{success}} = \prod_{\ell=1}^{L} \big(1 - P_e^{(\ell)}\big) > 1 - \varepsilon, \tag{26}$$

where $P_e^{(\ell)}$ is the per-layer error probability at layer ℓ, with sector width $\theta_\ell = \theta_0/(N-1)^{\ell-1}$, assuming $N_\ell = N$ is fixed across layers. At each layer, the new sector is the HPBW of the selected beam from the previous layer, which equals $\theta_{\ell-1}/(N-1)$ under the boundary-centered convention. Substituting (25) and using $1 - x \lesssim e^{-x}$ for small $x \geq 0$:

$$\prod_{\ell=1}^{L} \big(1 - P_e^{(\ell)}\big) \;\lesssim\; \exp\left(-\sum_{\ell=1}^{L} P_e^{(\ell)}\right) \;=\; \exp\left(-\frac{(K+1)e^{-K}\, H_{N-1}\, \theta_0}{101.5^\circ \cdot \text{SNR}} \sum_{\ell=1}^{L} \frac{1}{(N-1)^\ell}\right), \tag{27}$$

where $G^{(\ell)} = 101.5^\circ (N-1)^\ell/\theta_0$ grows geometrically with depth (sector narrows as $\theta_\ell = \theta_0/(N-1)^{\ell-1}$), provided $G^{(\ell)} \leq N_{\text{ant}}$ at every layer. The geometric sum evaluates to

$$\sum_{\ell=1}^{L} \frac{1}{(N-1)^\ell} = \begin{cases} L & N = 2, \\ \dfrac{1}{N-2}\left(1 - \dfrac{1}{(N-1)^L}\right) & N \geq 3, \end{cases} \tag{28}$$

converging to $1/(N-2)$ for $N \geq 3$ as $L \to \infty$, but growing linearly in L for $N = 2$ (no convergence). Since $\prod_\ell (1 - P_e^{(\ell)}) \leq \exp(-\sum_\ell P_e^{(\ell)})$, a *sufficient* condition for $P_{\text{success}} > 1 - \varepsilon$ is

$$\frac{(K+1)e^{-K}\, H_{N-1}\, \theta_0}{101.5^\circ \cdot \text{SNR}} \cdot \sum_{\ell=1}^{L} \frac{1}{(N-1)^\ell} \;\leq\; \ln \frac{1}{1-\varepsilon} \;\approx\; \varepsilon \quad (\varepsilon \ll 1), \tag{29}$$

with the sum given by (28). The constraint is sufficient but not necessary: satisfying it guarantees $P_{\text{success}} > 1 - \varepsilon$, but the true feasibility boundary is slightly looser.

Proposition 3. ***Total error probability constraint*** *A* sufficient *condition for* $P_{\text{success}} > 1 - \varepsilon$ *is*

$$\frac{(K+1)e^{-K}\, H_{N-1}\, \theta_0}{101.5^\circ \cdot SNR} \cdot \frac{1}{N-2}\left(1 - \frac{1}{(N-1)^L}\right) \;\leq\; \varepsilon, \quad N \geq 3, \varepsilon \ll 1, \tag{30}$$

which is a closed-form constraint on (L, N, SNR, θ_0) *suitable for further resource minimization.*

Remark. The approximation (25) requires $G(N, \theta) \cdot \text{SNR} \gg K + 1$. The resource-optimal beam count *without misdetection probability constraints* is $N^* = X^{1/\ln X} \approx 2.7$ for typical codebook sizes. At $\theta = 60^\circ$, this gives $G \approx 101.5^\circ \cdot 1.7/60^\circ \approx 2.9$, so $G \cdot \text{SNR} \approx 5\,\text{dB}$, which is marginal. Equation (25) and the constraint (30) should therefore be understood as providing the correct *functional form* of the reliability constraint. They identify the relevant parameters and their roles, but not always a tight numerical bound near N^*. For numerical evaluation of P_e at small N, the exact formula (16) must be used.

3.3 Error probability approximation quality

Table 1 compares (25) against the exact formula (16) at $K = 10$, SNR $= 5\,\mathrm{dB}$, $\theta = 60°$, $N_{\mathrm{ant}} = 128$, so $G(N, 60°) = \min(128,\ 101.5°(N-1)/60°)$. The harmonic approximation consistently underestimates P_e due to the neglected second-order terms $O\big((K+1)^2/(mG{\cdot}\mathrm{SNR})^2\big)$ in (20)–(21). The underestimate means the constraint (29) is optimistic by the same factor; the exact formula (16) should be used for tight numerical bounds.

Table 1: Harmonic approximation (25) vs. exact binomial sum (16). $K = 10$, SNR $= 20\,\mathrm{dB}$, $N_{\mathrm{ant}} = 128$. Values for $N \geq 48$ are omitted: the alternating binomial sum loses all significant digits in double precision (condition number $\sim 10^{11}$ at $N = 48$).

			P_e		
N	G	$G \cdot \mathrm{SNR}$ [dB]	Exact (16)	Harmonic (25)	Rel. error
$\theta_0 = 120°$					
2	0.85	19.3	3.69×10^{-5}	5.90×10^{-6}	-84%
4	2.54	24.0	8.03×10^{-6}	3.61×10^{-6}	-55%
8	5.92	27.7	3.29×10^{-6}	2.19×10^{-6}	-34%
16	12.69	31.0	1.63×10^{-6}	1.31×10^{-6}	-20%
32	26.22	34.2	8.64×10^{-7}	7.67×10^{-7}	-11%
$\theta_0 = 60°$					
2	1.69	22.3	8.26×10^{-6}	2.95×10^{-6}	-64%
4	5.08	27.1	2.75×10^{-6}	1.80×10^{-6}	-34%
8	11.84	30.7	1.35×10^{-6}	1.09×10^{-6}	-19%
16	25.38	34.0	7.30×10^{-7}	6.53×10^{-7}	-11%
32	52.44	37.2	4.07×10^{-7}	3.84×10^{-7}	-6%

Notation summary

N	number of beams per layer
N_{ant}	total number of physical ULA elements
θ	sector angular width (radians)
K	Rician K-factor (LOS-to-scatter power ratio)
P_{rx}	mean received power (before beamforming gain)
N_0	effective noise power ($kT_0B \cdot 10^{N_F/10}$)
$\mathrm{SNR} = P_{\mathrm{rx}}/N_0$	per-element SNR (linear)
$G(N,\theta) = \min(N_{\mathrm{ant}},\ 101.5°(N-1)/\theta)$	beamforming gain on correct beam
$n(N,\theta) = 101.5°(N-1)/\theta$	number of active elements per beam
$N_{\max}(\theta) = \lfloor N_{\mathrm{ant}}\theta/101.5° \rfloor + 1$	maximum feasible beams over sector
$\nu = GP_{\mathrm{rx}}K/(K+1)$	LOS noncentrality power
$\sigma^2_{\mathrm{tot}} = GP_{\mathrm{rx}}/(K+1) + N_0$	total variance per complex dimension
$H_n = \sum_{k=1}^{n} 1/k$	n-th harmonic number
$\gamma_E \approx 0.5772$	Euler–Mascheroni constant
$I_0(\cdot)$	modified Bessel function, first kind, order 0
$\mathcal{L}_X(s) = \mathbb{E}[e^{-sX}]$	Laplace transform of RV X